

SDOC Check

Shipping document verification, with the evidence on screen.

sdoc-hackathon.vercel.app

works on a phone · no login · no install

The problem

Shipping teams receive hundreds of emails. Some ask them to check a **Shipping Instruction (SI)** against the **draft Bill of Lading (BL)** before it is finalised. Doing that by eye is slow, and a wrong port, weight or company name is expensive.

What SDOC Check does

It reads the inbox, finds those requests, compares the two documents on seven fields and shows every difference with **the exact line it came from**. Anything it cannot decide goes to a person. It never guesses.

The seven fields it compares

Shipper

Consignee

Notify party

Port of loading

Port of discharge

Container count

Gross weight

520

emails in the demo inbox

220

document checks

63

clean

46

mismatches

20

need a person

91

waiting for the draft BL

Inbox520

AI-assisted3

Document checks

Needs review19

Mismatches46

No mismatch63

All document checks220

Other mail

New SI requests125

Invoice queries75

General60

Junk40

From an email to a verdict, in six steps

1 Sort the inbox RULES

Every email is filed as a document check, a new SI request, an invoice query, general mail or junk, using keywords. If the rules are unsure, Gemini (AI) may suggest a category.

2 Read the documents RULES

Text, PDF, Word and Excel files are turned into text. A scan with no text, or a broken file, is never guessed at: it is flagged.

3 Find the seven fields RULES AI BACKUP

Values are found whatever the wording ("Port of Loading", "POL", "Load Port"). Each one keeps the exact line it was read from. If a value is missing, Gemini may look for it, but only counts if it quotes a real line of the document.

4 Compare RULES

Formatting is tidied first: company endings (LTD = LIMITED), port codes, container size and type, and weights in kilograms, tonnes or pounds. Then each field is compared. **The AI never decides whether two values match.**

5 Give a verdict

OK

Every field agrees.

MISMATCH

At least one field differs. The fields are listed with both values.

NEEDS REVIEW

Wrong document, missing attachment, unreadable scan or a blank value.

AWAITING

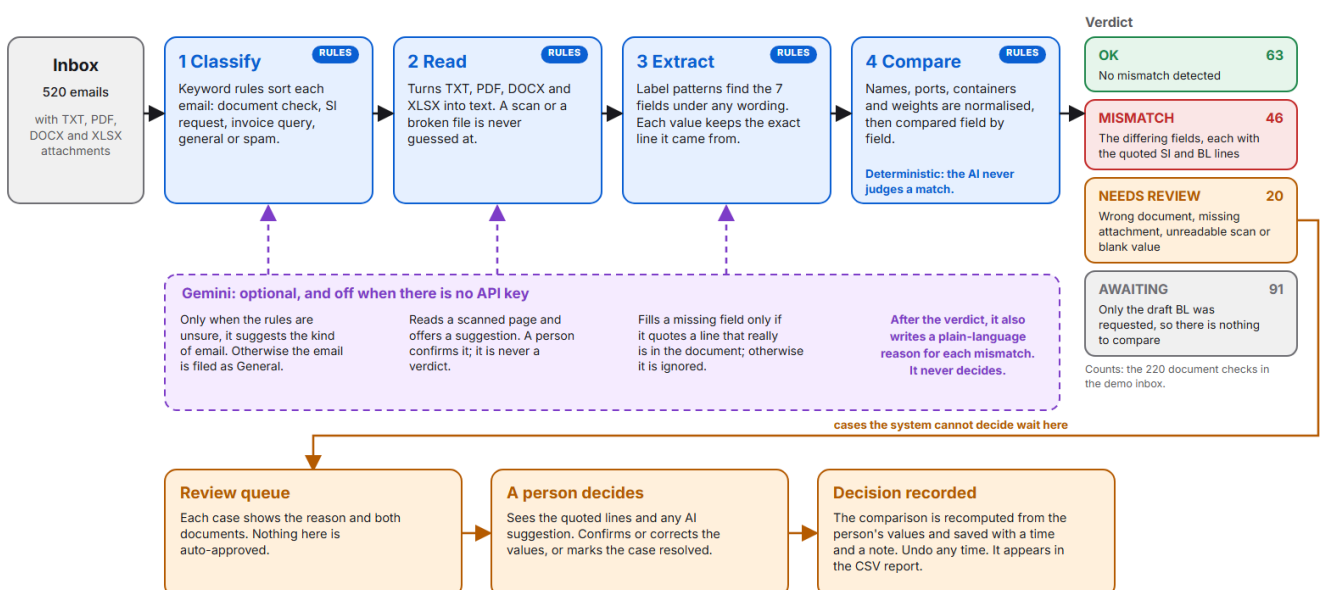
Only the draft BL was requested, so there is nothing to compare yet.

6 A person settles what is uncertain PERSON

Uncertain cases wait in **Needs review** with both documents, the reason and the quoted lines. A person confirms or corrects the values, the report updates, and the decision is saved.

SDOC Check: from an email to a verdict

Rules decide. AI only helps where rules cannot, and a person settles anything uncertain.



□ Rules: deterministic and reproducible

□ AI: optional, always checked or confirmed

□ A person

Every value on screen links back to its source line, file and line number.

What you can do with it

See the proof

Every value shows the **exact line** of the source document, with the file name and line number. A person can check it at a glance.

Review queue

Edit values, add a note, then **Save & re-check** or **Mark as resolved** . **Undo** reverses it, and **Retry** runs one email again. Decisions are saved.

Find things fast

Filters (Needs review, Mismatches, No mismatch, AI-assisted), search, arrow-key browsing, and a direct link to any single email.

Discrepancy report

One click downloads a CSV of every document check: the result, the differing fields, whether a person reviewed it and whether AI helped.

AI help, clearly marked

Anything Gemini touched is tagged **AI-assisted** , carries a disclaimer, and marks values it found with **AI** . Gemini also writes a plain-language reason for every mismatch and every case that needs review, and a grey note says when it was asked and found nothing.

Works anywhere

A live link, a phone layout, no install and no login. Results are ready instantly, and reviewer decisions survive a refresh.

CONFIRM THE VALUES

FIELD	SHIPPING INSTRUCTION read from the document	DRAFT BILL OF LADING read from the document
Shipper	APRIL FINE PAPER TRADING "Shipper (Principal or Seller): APRIL FINE PAPER TRADING" email_516_SI.txt - line 4	APRIL FINE PAPER TRADING "Shipper: APRIL FINE PAPER TRADING" email_516_BL.txt - line 4
Consignee	KPP-ANTALIS (SINGAPORE) PTE. LTD. "Consignee: KPP-ANTALIS (SINGAPORE) PTE. LTD." email_516_SI.txt - line 5	KPP-ANTALIS (SINGAPORE) PTE. LTD. "Consignee: KPP-ANTALIS (SINGAPORE) PTE. LTD." email_516_BL.txt - line 6
Notify party	KPP-ANTALIS (SINGAPORE) PTE. LTD. "Notify: KPP-ANTALIS (SINGAPORE) PTE. LTD." email_516_SI.txt - line 6	KPP-ANTALIS (SINGAPORE) PTE. LTD. "Notify Party/Intermediate Consignee: KPP-ANTALIS (SINGAPORE) PTE. LTD." email_516_BL.txt - line 8
Port of loading	NHAVA SHEVA, INDIA "PORT OF LOADING: NHAVA SHEVA, INDIA" email_516_SI.txt - line 7	NHAVA SHEVA, INDIA (INNSA) "Load Port: NHAVA SHEVA, INDIA (INNSA)" email_516_BL.txt - line 9
Port of discharge	CONAKRY, GUINEA "POD: CONAKRY, GUINEA" email_516_SI.txt - line 8	CONAKRY, GUINEA (GNCKY) "POD: CONAKRY, GUINEA (GNCKY)" email_516_BL.txt - line 10
Container count	10 × 20'FCL "No. of Containers or Packages: 10 × 20'FCL" email_516_SI.txt - line 9	10 × 20'FCL "Container Count: 10 × 20'FCL" email_516_BL.txt - line 11
Gross weight (kg)	Enter value "Gross Weight毛重(KGS): N/A" email_516_SI.txt - line 10	235,550 KG "Gross Weight (KG): 235,550 KG" email_516_BL.txt - line 12

Reviewer note (optional)

Save & re-check

Mark as resolved

Re-check compares your values and updates the report.

A case that needs review

A blank value on the SI side. The person fills it in, adds a note and saves.

The comparison is **recomputed from what they entered**, the case is marked as verified by a person, and it can be undone.

Under every box is the line of the document the value came from, so the reviewer never has to open the file to check.

How we know it works

We tried to break it

Real documents were changed thousands of ways, then checked again.

- **1,274 of 1,274** harmless changes (capital letters, punctuation, kilograms vs tonnes) raised no false alarm.
- **1,572 of 1,572** real defects were caught.
- **988 of 988** blanked values went to a person.
- **267 of 267** unfamiliar wordings were escalated, never passed silently.

A real bug, found and fixed

Weights exactly 1 kg apart were counted as equal. The tests found it, it is fixed, and a test now guards it.

Also

- 100% on the organisers' scorer, on their data and three fresh datasets. It is synthetic data, so the fuzzing is the harder test.
- 120 automated tests.
- The AI readings of the three scanned emails were checked against the pages by hand: 21 of 21 values correct.

Honest limits

- All data is synthetic, and there is no login.
- The email-sorting keywords were written from this dataset's wording; a real inbox would need them widened.
- On scanned emails only the SI page is read by Gemini.
- Only cases that need review can be edited.

Where it runs

- **Vercel**: the web app
- **Supabase**: saved reviewer decisions
- **Gemini**: optional AI help
- **GitHub**: code; pushing to main deploys

Try it in two minutes

Search for an email id on the live site, or add it to the address, for example sdoc-hackathon.vercel.app/#email_004.

email_001

A clean check: "No mismatch detected"

email_004

Two mismatches, each with the exact SI and BL line it was read from

email_003

Only the draft BL was requested: nothing to compare, and it says so

email_516

A blank value: fill it in, Save & re-check, then Undo

email_501

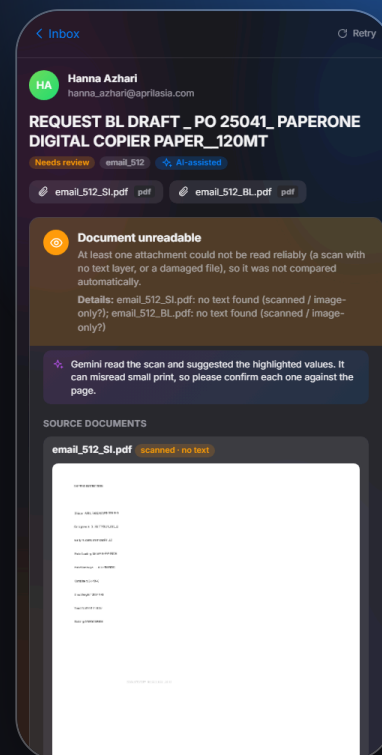
The wrong document: the "BL" attachment is really an invoice

email_512 to 514

Scanned pages read by Gemini as a suggestion, tagged AI-assisted

email_519, 520

Gemini was asked and found nothing, so a grey note says so



An AI-read scan on a phone

sdoc-hackathon.vercel.app

Code, docs and the validation report: github.com/Tynashe13/sdoc-hackathon